

# Conceptual Design for a BSL-3 Suitable Stage-Top Incubator



**Prepared by:**  
Joab Gray Kloppers

**Prepared for:**  
Mandy Mason  
Robyn Verrinder  
Justin Pead  
University of Cape Town

February 3, 2026

# Contents

|          |                                   |           |
|----------|-----------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>               | <b>1</b>  |
| 1.1      | Background and Summary . . . . .  | 1         |
| 1.2      | Scope and Limitations . . . . .   | 1         |
| <b>2</b> | <b>Research</b>                   | <b>2</b>  |
| 2.1      | Zeis/Pecon . . . . .              | 2         |
| 2.2      | Ibidi . . . . .                   | 3         |
| 2.3      | Tokai-Hit . . . . .               | 4         |
| 2.4      | Oko-Lab . . . . .                 | 4         |
| 2.5      | Shortfalls . . . . .              | 5         |
| <b>3</b> | <b>Feasibility Analysis</b>       | <b>6</b>  |
| 3.1      | Commercial Assessment . . . . .   | 6         |
| 3.1.1    | Pecon/Zeis . . . . .              | 6         |
| 3.1.2    | ibidi . . . . .                   | 6         |
| 3.1.3    | Tokai Hit . . . . .               | 6         |
| 3.1.4    | Oko Lab . . . . .                 | 6         |
| 3.2      | Requirements . . . . .            | 6         |
| 3.3      | Specifications . . . . .          | 6         |
| 3.4      | Feasibility . . . . .             | 6         |
| <b>4</b> | <b>Conceptual Design</b>          | <b>7</b>  |
| 4.1      | Design Iterations . . . . .       | 7         |
| 4.2      | Final Conceptual Design . . . . . | 7         |
| <b>5</b> | <b>Discussion and Results</b>     | <b>8</b>  |
| <b>6</b> | <b>Conclusion</b>                 | <b>9</b>  |
| 6.1      | Recommendations . . . . .         | 9         |
|          | <b>Bibliography</b>               | <b>10</b> |

# Chapter 1

## Introduction

### 1.1 Background and Summary

Long term Live-Cell imaging is a very useful research tool when determining long term behavior of cell cultures in various situations. Commercial solutions are unable to provide adequate containment for the use within a Bio-Safety Level 3 (BSL-3) and often require the use of a separate containment enclosure, which introduces additional costs and complexity.

This project focuses on the conceptual design of a low-cost Stage-Top Incubator that is capable of being utilized within a BSL-3 laboratory, without the requirement for additional containment.

### 1.2 Scope and Limitations

This project is limited to the following:

- Preliminary Research
- Temperature Control
- Compatibility with Multi-Well microtiter plates

This project is limited to compatibility with a BSL-3 Lab and does not consider higher containment requirements. The environmental control aspects of the incubator are limited to just temperature control for the preliminary conceptual design. The initial design is limited to compatibility with a multi-well microtiter plate.

# Chapter 2

## Research

The conceptual design began with assessing existing commercial products to identify the key environmental control specifications required for the incubator to function correctly. Additionally, this was useful in identifying why the commercial products are incompatible with a BSL-3 lab when used without a separate external biological containment mechanism.

### 2.1 Zeis/Pecon

The Zeis/Pecon stage top incubator utilities four modules to control the environmental conditions within the incubator stage.

The Pecon Temperature Module S[1] regulates the temperature of the incubator. It has four 8-pin connectors that are used for temperature control of heated components such as the Incubator lid, incubator base, microscope objective and humidifier device. The temperature module is the base module for the entire incubation system, providing electrical power and communications to all connected devices. It directly connects to the CO<sub>2</sub> module to provide power, communication and makes a connection to the gas circulation system to heat the gas flow provided to the incubation chamber. The temperature module requires an external device such as a PC or compatible microscope for operation as it does not have any control panel. The module maintains culture temperature by monitoring and controlling the temperature of the incubator base, lid and optionally the microscope objective. The module infers the culture temperature using predefined temperature relations, or by performing a pre-experiment calibration whereby an identical set-up is placed in the chamber with the included culture temperature sensor to tune the required chamber temperatures.

The Pecon CO<sub>2</sub> Module S[2] attaches to the temperature module for electrical power and communication, and provides CO<sub>2</sub> concentration control to the airflow passed into the incubation chamber. The CO<sub>2</sub> module circulates gas throughout all the connected units and mixes the gas before pumping the resultant mixture into the incubation chamber. The module contains a CO<sub>2</sub> sensor which monitors the concentration of CO<sub>2</sub> in the gas mixture, and controls the release of pure CO<sub>2</sub> into the mixing chamber in order to maintain the desired CO<sub>2</sub> concentration. Due to the method of operation, it requires a >95% purity CO<sub>2</sub> supply.

The Pecon O<sub>2</sub> Module S[3] attaches to the CO<sub>2</sub> module for electrical power and communication. This module functions similarly to the CO<sub>2</sub> module, however it regulates the concentration of O<sub>2</sub> present in the gas mixture by controlling the release of a pure N<sub>2</sub> supply into the air mixture to displace the O<sub>2</sub> present. This method of O<sub>2</sub> displacements means that the O<sub>2</sub> module is only capable of providing hypoxia conditions by lowering O<sub>2</sub> concentration. Due to the method of operation, it requires a >99% purity N<sub>2</sub> supply.

Both the CO<sub>2</sub> module and O<sub>2</sub> module are supplied with a humidifying bottle which functions by bubbling the the gas airflow thorough distilled water to raise the relative humidity (RH). The Pecon Heating Device Humidity S1[4] is an optional module which connects to the Temperature module to heat the humidifying bottle to increase the RH and maintain adequate gas temperature provided to the incubation chamber

The incubator stage includes a heated glass topped lid[5] and a heated insert which accepts mounts for the different culture vessels used in microscopy. The heated insert directly heats the culture medium through physical contact, and the heated lid provides radiative heating and condensation prevention by heating the glass panel directly. The insert supports perfusion experiments utilising two ports which allow tubes to be run to the culture.

## 2.2 Ibidi

The Ibidi stage top incubator functions similarly to the Zeis/Pecon module, however it makes use of only 3 modules to maintain environmental conditions.

The Heating System Silver Line[6] module regulates the temperature of the incubator. It makes use of 6 independent channels connected via 4 ports, two of which connect to two channels each. These channels are used to monitor and control the temperature of the components used in the system, such as the heated lid and heated plate. The module maintains culture temperature by monitoring and controlling the temperature of the heated lid, heated base and optionally the microscope objective. The module infers the culture temperature using predefined temperature relations, or by performing a calibration whereby an identical set-up is placed in the chamber with the included thermocouple to tune the required chamber settings to meet the desired culture temperature. The temperature module can be operated using the on device buttons and screen, or it can be connected to a PC for control using the free software. The temperature module does not heat the gas mixture.

The Gas Incubation System comes in two different models, the CO<sub>2</sub> - Silver Line and the CO<sub>2</sub>/O<sub>2</sub> - Silver Line. The difference between these two systems is the former is only capable of regulating CO<sub>2</sub> concentration, where as the later has the ability to regulate O<sub>2</sub> concentration as well as CO<sub>2</sub> concentration. The gas incubation system is responsible for heating the gas mixture and the humidifying column to the appropriate temperature.

The Humidifying system works by mixing airflow directly from the Mixing module and from the humidifying column, which bubbles a separate airflow through heated water to obtain approximately 100% RH air. The gas mixer module then controls the mix ratio of dry gas to 100% RH gas to obtain a controlled humidity flowing into the incubation chamber.

The Ibidi incubation chamber comes in two versions, one version supports culture mediums with Petri dishes and slides, the second version supports multi-well plates. The Petri dish/Slide version of the chamber supports perfusion experiments using two ports which allow tubes to be run to the culture. Both versions support direct medium temperature monitoring which allow for more accurate temperature control by directly measuring the temperature of the culture medium.

The Ibidi system functions in a very similar fashion to the Pecon system, with the addition of being

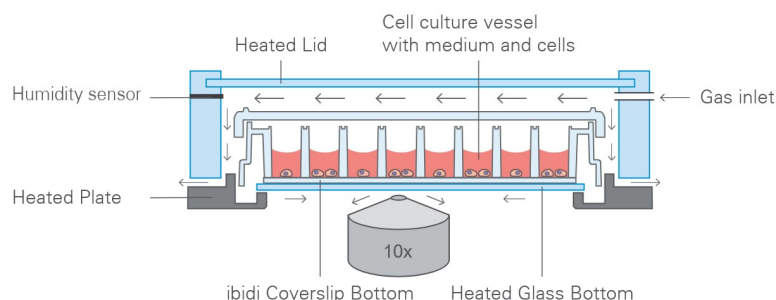


Figure 2.1: Diagram Showing the Airflow Path Within an Ibidi Multi-Well Incubator Chamber [6]

able to control the relative humidity in the chamber, rather than a forced approximate 100% relative humidity. This system still poses the same problem wherein the airflow is designed to passively exhaust from the incubation chamber, as shown in Figure 2.1, and therefore does not have a seal that meets BSL-3 containment standards.

## 2.3 Tokai-Hit

The Tokai-Hit system functions similarly to the Pecon and Ibidi systems however it utilises a single module to regulate the environmental conditions for temperature and CO<sub>2</sub>. Humidity is controlled by a heated water bath inside the chamber, and O<sub>2</sub> control is achieved by utilising an optional module which then overrides all gas control.

The Tokai-Hit STXG/STXF temperature and CO<sub>2</sub> controller regulates both temperature and CO<sub>2</sub> concentration in the incubation chamber. The controller is capable of regulating the temperature of the top-lid heater, water bath heater, stage heater and the microscope objective heater. Additionally it is capable of regulating the airflow into the chamber depending on the model and air supply.

The incubator chamber uses a single design that is compatible with all major culture dishes including Petri dishes, slides and multi-well plates. This chamber supports perfusion experiments, drug administration and direct culture temperature measurement via access ports designed into the lid.

This system still poses the same issue as the aforementioned chambers, which prevents this chamber from forming a completely sealed environment suitable for a BSL-3 lab.

## 2.4 Oko-Lab

The Oko-Lab utilises three modules to regulate environmental conditions within the incubation chamber

The Oko-Lab H301-T-UNIT-BL-PLUS [8] unit is responsible for maintaining the thermal conditions within the chamber enclosure. It operates by monitoring and controlling the temperature of the chamber base, lid and optionally the microscope objective. This in turn heats the sample to the desired temperature. This heating unit is capable of sample temperature feedback utilising a thin thermocouple that can be placed in distilled water near the sample to obtain more accurate sample temperature control. Otherwise the unit is capable of using pre-sets/calibrations to infer sample temperature based on enclosure temperatures. The heating unit can be operated via a computer using available software

or using optional touch screen accessory. The heating unit is capable of interfacing with the Oko-Lab Bold Line Gas Controllers.

## **2.5 Shortfalls**

These dont work for these reasons

# Chapter 3

## Feasibility Analysis

### 3.1 Commercial Assessment

Asses commercial solutions and provide technical specifications for the,

#### 3.1.1 Pecon/Zeis

#### 3.1.2 ibidi

#### 3.1.3 Tokai Hit

#### 3.1.4 Oko Lab

### 3.2 Requirements

Describe requirements needed for this project

### 3.3 Specifications

Describe the specifications that are decided on for this project.

### 3.4 Feasibility

discus whether undertaking this project is feasible.



## Chapter 4

# Conceptual Design

### 4.1 Design Iterations

These were the iterations and steps taken when conceptually designing the product

### 4.2 Final Conceptual Design

The final design is the following, and uses the following features.

## Chapter 5

# Discussion and Results

# Chapter 6

## Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

### 6.1 Recommendations

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

# Bibliography

- [1] Pecon, “Pecon/Zeis TempModule S Manual.”
- [2] —, “Pecon/Zeis CO2 Module S Manual.”
- [3] —, “Pecon/Zeis O2 Module S Manual.”
- [4] —, “Pecon/Zeis Heating Device Humidity S1 Manual.”
- [5] —, “Incubator P S1 compact Manual.”
- [6] ibidi, “ibidi Heating System Multiwell Plate - Silver Line Manual.” [Online]. Available: [https://ibidi.com/img/cms/downloads/in/IN\\_12150\\_HeatingSystem\\_Multiwell\\_SL.pdf](https://ibidi.com/img/cms/downloads/in/IN_12150_HeatingSystem_Multiwell_SL.pdf)
- [7] “O-Ring-Handbook.” [Online]. Available: [https://www.parker.com/content/dam/Parker-com/Literature/Praedifa/Catalogs/Catalog\\_O-Ring-Handbook\\_PTD5705-EN.pdf](https://www.parker.com/content/dam/Parker-com/Literature/Praedifa/Catalogs/Catalog_O-Ring-Handbook_PTD5705-EN.pdf)
- [8] Oko-Lab, “H301-T-UNIT-BL-PLUS\_manual.” [Online]. Available: [https://www.oko-lab.com/images/okaImages/PDF-files/H301-T-UNIT-BL-PLUS\\_Manual.pdf](https://www.oko-lab.com/images/okaImages/PDF-files/H301-T-UNIT-BL-PLUS_Manual.pdf)